CLINICAL · LARYNX & AIRWAY

Laryngotracheal trauma and stenosis

A blow to the front of the neck can break the larynx without breaking the skin. Nearly half of those with a laryngeal fracture die at the scene; of the rest, many look well for a few hours while oedema and haematoma quietly close the airway. Secure the airway without a tube that may make a false passage, then restore the framework early, before scar turns a repairable injury into a lifelong stenosis.

AT A GLANCE

- Loss of thyroid prominence, emphysema and crepitus.
- Tracheostomy under local anaesthesia, not blind intubation.
- Repair displaced fractures and tears within days.
- Intubation is the commonest cause of subglottic stenosis.
- Penetrating neck injury: hard signs mean theatre; otherwise CT angiography and selective exploration.

1 CHAPTER ONE Overview

Blunt, penetrating and from within.

1.1 Causes

- **Blunt:** road traffic accidents (the neck striking the steering wheel or dashboard with the head extended), blows and kicks, clothesline injuries from a stretched wire or cable, and strangulation.
- **Penetrating:** knives, glass and gunshot wounds.
- **Internal (iatrogenic):** intubation, endoscopy, inhalation and caustic burns, and foreign bodies.

1.2 Why the larynx is usually spared

It is shielded by the mandible above, the sternum below and the spine behind, and it is mobile. Fractures need a direct anterior blow with the head extended, and become commoner after forty, when the cartilages calcify and become brittle; in

children the elastic cartilages bend rather than break, but soft tissue injury and airway swelling are more dangerous.

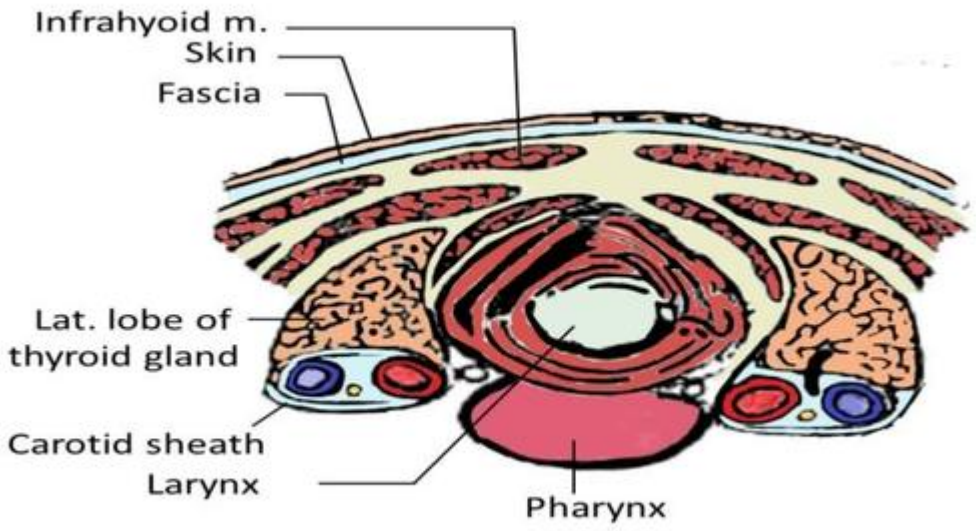


FIGURE 1 The relations of the larynx in cross-section.
Anatomy atlas · Doss p. 170

2 CHAPTER TWO Pathology

What breaks.

- Haematoma and oedema of the supraglottis or subglottis.
- Mucosal tears of the larynx or pharynx, with surgical emphysema.
- Dislocation or avulsion of an arytenoid at the cricoarytenoid joint.
- Dislocation of the cricothyroid joint, injuring the recurrent laryngeal nerve that runs just behind it.
- Fracture of the hyoid.
- Fractures of the thyroid cartilage, vertical or transverse: an upper fracture can avulse the epiglottis and false folds; a lower one displaces or tears the true folds and anterior commissure.
- Fracture of the cricoid, the ring that holds the airway open.
- Fractures of the upper tracheal rings.

- *Laryngotracheal separation*: the trachea parts from the cricoid and retracts into the mediastinum, usually with bilateral recurrent nerve injury.

THE LUCID INTERVAL

A patient with a laryngeal fracture may talk and breathe normally on arrival, then obstruct suddenly hours later as haematoma and oedema accumulate. Every significant anterior neck injury is observed in hospital with the airway assessed by endoscopy.

3 CHAPTER THREE Clinical features

Voice, breath, swallow, blood.

3.1 Symptoms

- Respiratory distress and stridor.
- Hoarseness or aphonia.
- Painful, difficult swallowing, with aspiration.
- Pain in the larynx, worse with speaking and swallowing.
- Haemoptysis from mucosal tears.

3.2 Signs

- Bruising and abrasions of the neck.
- Tenderness over the larynx.
- *Surgical emphysema*, increasing with coughing.
- Loss of the thyroid prominence and flattening of the neck contour; the thyroid notch may be impalpable.
- A palpable step or gap between fragments, and crepitus.
- Separation of the cricoid from the larynx or trachea.

3.3 Associated injuries

Cervical spine (assume it is injured), great vessels, oesophagus and pharynx, chest

(pneumothorax), head and abdomen.

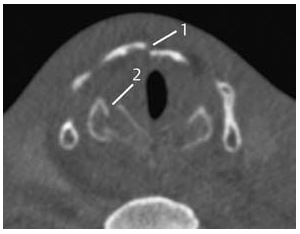
4 CHAPTER FOUR Evaluation

Only when the airway allows.

- **Flexible nasendoscopy:** the most useful first examination in a stable patient: oedema, haematoma, lacerations, exposed cartilage, arytenoid displacement, vocal fold movement, and the size of the airway.
- **CT of the larynx (with 3D reconstruction):** cartilage fractures, joint dislocation, displacement; also the cervical spine and vessels. Only when the airway is safe.
- **Rigid endoscopy** (laryngoscopy, oesophagoscopy, bronchoscopy) under anaesthesia, after the airway has been secured, to map injuries.
- **Chest radiograph** for pneumothorax and mediastinal air; **contrast swallow** for pharyngo-oesophageal tears; **angiography** if vascular injury is suspected.

IMAGING IN THE APP 1 case

Laryngeal trauma on imaging.



Laryngeal Trauma

Trauma · 10 images

SCHAEFER–FUHRMAN GRADING OF LARYNGEAL INJURY

GROUP	FINDINGS	MANAGEMENT
1	Minor haematoma or laceration, no fracture	Observe, humidify, steroids
2	Oedema, haematoma, minor mucosal disruption, no exposed cartilage, non-displaced fracture	Observe; tracheostomy if the airway is threatened
3	Massive oedema, mucosal tears, exposed cartilage, displaced fracture, cord immobility	Tracheostomy and open repair

GROUP	FINDINGS	MANAGEMENT
4	As group 3 with a disrupted anterior commissure or unstable framework	Tracheostomy, open repair and stenting
5	Complete laryngotracheal separation	Tracheostomy and repair by anastomosis

5 CHAPTER FIVE Management

Airway, then framework.

5.1 The airway

- **Acute distress:** awake *tracheostomy under local anaesthesia*, then laryngoscopy under general anaesthesia. Orotracheal intubation is risky: a tube can enter a false passage through torn mucosa or complete a laryngotracheal separation. If intubation is attempted, it is by the most experienced operator, fibreoptically and with the neck prepared for a surgical airway.
- **Stable airway:** flexible endoscopy, then CT.

5.2 Conservative treatment

For minor injuries with a normal or near-normal endoscopy and no displaced fracture: admission and observation, voice rest, humidified air, head elevation, systemic steroids early and in full dose (to reduce oedema and scarring), antibiotics (to prevent perichondritis), and anti-reflux treatment.

5.3 Surgical repair

Indicated for displaced or unstable fractures, exposed cartilage, large mucosal tears, disruption of the anterior commissure, arytenoid dislocation and cord immobility, and laryngotracheal separation. Repair early, within the first few days and ideally before ten days, before granulation and scar form.

- **Exposure:** a midline thyrotomy (laryngofissure) to inspect and close the mucosa.

- **Mucosa:** all lacerations sutured; exposed cartilage covered with local flaps or grafts; loose fragments removed.
- **Framework:** fractures reduced and fixed with wire or titanium miniplates.
- **Epiglottis:** reattached, or excised if avulsed.
- **Arytenoid:** reduced if dislocated; removed if completely avulsed.
- **Anterior commissure:** a silastic keel prevents a web.
- **Stents:** a soft laryngeal stent or silicone tube for comminuted fractures, left for two to six weeks; stents themselves provoke granulation.
- **Laryngotracheal separation:** end-to-end anastomosis, with assessment of both recurrent nerves.

OPERATIVE GUIDES IN THEATRE 1 operation

Repair of laryngeal trauma.

Laryngeal Trauma
Laryngeal trauma

5.4 Complications

Laryngeal stenosis (supraglottic, glottic or subglottic), perichondritis and abscess, chondronecrosis, vocal fold paralysis or fixation, webs, and a poor voice.

6 CHAPTER SIX Laryngotracheal stenosis

The late result.

6.1 Causes

Prolonged or traumatic *intubation* (the commonest cause: pressure necrosis of the posterior glottis and subglottis by the tube and cuff), external trauma, high tracheostomy and cricothyroidotomy, burns and caustics, radiotherapy, granulomatosis with polyangiitis, relapsing polychondritis, sarcoidosis, reflux, and idiopathic subglottic stenosis in women.

6.2 Presentation

Progressive dyspnoea and stridor, often biphasic, sometimes with hoarseness; many patients are misdiagnosed with asthma.

6.3 Evaluation

History of intubation or injury; office endoscopy of the supraglottis and glottis and vocal fold movement; CT of the subglottis and trachea; and direct laryngoscopy and bronchoscopy to measure the length, grade and site and to palpate the arytenoids (fixation versus paralysis).

COTTON–MYER GRADING OF SUBGLOTTIC STENOSIS

Grade I, up to 50% obstruction; grade II, 51 to 70%; grade III, 71 to 99%; grade IV, no detectable lumen.

6.4 Treatment

- **Supraglottic:** endoscopic excision of scar; recurrence is common after radiotherapy.
- **Glottic:** an anterior web affects voice more than breathing and is divided with a keel; posterior glottic scarring that fixes the folds is treated by division of the interarytenoid scar, or by procedures that enlarge the glottis at the expense of voice (as for bilateral paralysis).
- **Subglottic:** endoscopic radial incisions with laser and balloon dilatation (with topical mitomycin or steroid injection) for thin, soft stenoses; *laryngotracheal reconstruction* with cartilage grafts, or *cricotracheal resection* with anastomosis, for firm, long or high-grade stenoses.
- **Tracheal:** resection and end-to-end anastomosis is definitive; balloon dilatation for thin webs or unfit patients.
- **Complex, multilevel stenosis:** a Montgomery T-tube stents the airway while preserving voice. Expanding metal stents are avoided in benign disease because of granulation and erosion.

OPERATIVE GUIDES IN THEATRE 4 operations

Acquired laryngeal stenosis, tracheal stenosis, laryngotracheal reconstruction and tracheal resection.

Acquired Laryngeal Stenosis
Acquired laryngeal stenosis
Management of Tracheal Stenosis
Tracheal stenosis
Laryngotracheal
Laryngotracheal reconstruction
Tracheal Resection
Tracheal resection

6.5 Prevention

Use the right-sized tube, keep cuff pressure below about 25 to 30 cm of water, avoid prolonged translaryngeal intubation where tracheostomy is better, and avoid placing a tracheostomy through the first ring or cricoid.

PENETRATING NECK INJURY (OXFORD TEXTBOOK)

Penetrating injuries are those that breach the **platysma**; they carry a mortality of up to about 6%, and half of deaths are from bleeding. The anterior neck is divided into three zones. **Zone I** runs from the clavicles and sternal notch to the cricoid and contains the great vessels at the thoracic inlet, trachea, oesophagus, lung apices and thoracic duct. **Zone II**, from the cricoid to the angle of the mandible, is the commonest site and the easiest to expose, with the best prognosis. **Zone III**, from the angle of the mandible to the skull base, contains the distal carotid and vertebral arteries and cranial nerves IX to XII and is hard to reach surgically.

Management follows trauma life support principles, with control of the airway (often by early intubation or a surgical airway) and of bleeding by direct pressure. **Hard signs**, such as an expanding haematoma, active bleeding, air bubbling through the wound, shock not responding to fluids, absent pulse, bruit or evolving stroke, mean immediate exploration in theatre. Stable patients with **soft signs** (haemoptysis, dysphonia, dysphagia, surgical emphysema, a non-expanding haematoma) or none are investigated: **CT angiography** has replaced catheter angiography as the first test in all three zones, supplemented by endoscopy or contrast swallow for the aerodigestive tract. The old rule of mandatory exploration of every zone II wound has given way to this **selective approach**, which reduces negative explorations. Carotid injuries found at exploration are generally repaired rather than ligated.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Mechanism** of laryngeal fracture and why it is commoner after forty.
- **Signs:** loss of thyroid prominence, emphysema, crepitus.
- **Airway first:** tracheostomy under local anaesthesia, not intubation.
- **Indications and timing** of open repair.
- **Laryngotracheal stenosis:** causes, grading and treatment.

7.2 Common pitfalls

COMMON PITFALLS

- Discharging a talking patient with a laryngeal injury who later obstructs.
- Intubating blindly through a disrupted larynx.
- Delaying repair beyond the first week or two.
- Treating post-intubation subglottic stenosis as asthma.

Written for this app from Oxford Textbook of Otolaryngology ch. 53, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 25, Dhingra ch. 57 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · TRAUMA

Laryngeal Trauma

IMAGING FINDINGS

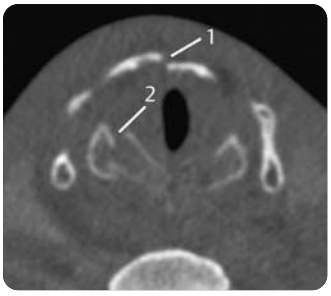
- Thyroid and cricoid cartilage fractures with displacement
- Arytenoid dislocation
- Haematoma and air narrowing the airway

TOP DIFFERENTIALS

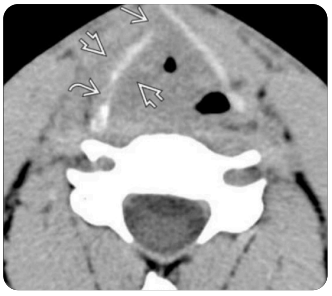
- Cord paralysis
- Unossified cartilage mimicking a fracture

CHECKLIST

- Secure airway before scanning
- Cricoid fractures and displaced fragments usually need repair



CT, axial bone window. CT at the level of the glottis showed multiple fractures; this figure shows a midline fracture with slight inward…



Axial NECT through the supraglottic larynx shows step-off at the superior thyroid notch. There is a subtle fracture of the right thyroid lamina…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 278.